Emanuele Guidotti

Swiss National Science Foundation (SNSF) Ambizione Fellow
University of Lugano (USI) and Lake Lucerne Institute (LLUI)
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EMPLOYMENT AND EDUCATION

Swiss National Science Foundation (SNSF) Ambizione Fellow — 2025 — Now
- Institute of Finance, **University of Lugano**
- Department of Wealth Creation, **Lake Lucerne Institute**

Postdoc in Finance - **USI Lugano** — 2023 — 2025
Supervisor: L. Fresard

PhD in Finance, **University of Neuchâtel** — 2019 — 2023
Supervisor: T.A. Kroencke Co-supervisor: D. Ardia

MSc in Finance, **University of Milan** — 2015 — 2017
Final evaluation: *summa cum laude.* Supervisor: S.M. Iacus

BSc in Physics, **University of Milan** — 2012 — 2015
Final evaluation: *summa cum laude.* Supervisor: S.M. Iacus and A. Vicini

RESEARCH INTERESTS

Financial markets, Data science, Artificial intelligence.

ACADEMIC VISITS

VISITING SCHOLAR, **VU Amsterdam**, Department of Finance — Jan-Apr 2027
Host: Albert Menkveld

VISITING SCHOLAR, **Stockholm Business School**, Finance Section — Sept-Dec 2026
Host: Björn Hagströmer

VISITING SCHOLAR, **Robert H. Smith School of Business**, Finance Department — Apr-Jun 2026
Host: Albert S. (Pete) Kyle

VISITING SCHOLAR, **HEC Montréal**, Department of Decision Sciences — May 2022
Host: David Ardia

VISITING SCHOLAR, **University of Tokyo**, Mathematical Sciences — Four non-consecutive months between 2017 and 2018
Host: Nakahiro Yoshida

PUBLISHED PAPERS

"Efficient Estimation of Bid-Ask Spreads from Open, High, Low, and Close Prices"
D. Ardia, E. Guidotti, T.A. Kroencke
Journal of Financial Economics (2024), *161*, 103916 [link]

"Text Classification with Born's Rule"
E. Guidotti, A. Ferrara
Advances in Neural Information Processing Systems (2022), *35*, 30990-31001 [link]

"calculus: High-Dimensional Numerical and Symbolic Calculus in R"
E. Guidotti
Journal of Statistical Software (2022), *104*(5), 1-37 [link]

"A worldwide epidemiological database for COVID-19 at fine-grained spatial resolution"
E. Guidotti
Scientific Data (2022), *9*(1), 112 [link]

"COVID-19 Data Hub"
E. Guidotti, D. Ardia
Journal of Open Source Software (2020), *5*(51), 2376 [link]

"Are official confirmed cases and fatalities counts good enough to study the COVID–19 pandemic dynamics? A critical assessment through the case of Italy"
K. Bartoszek, E. Guidotti, S. Iacus, M. Okrój.
Nonlinear Dynamics (2020), *101*(3), 1951-1979 [link]

WORKING PAPERS

"A Theory of Price Formation in Competitive Financial Markets"
E. Guidotti
SSRN, preprint. [link]

"The Invisible Handshake: Persistent Overpricing by Adaptive Market Agents"
L. Foscari, E. Guidotti, N. Cesa-Bianchi, T. Chavdarova, A. Ferrara
arXiv, preprint. Accepted at the NeurIPS 2025 Workshop on Algorithmic Collective Action [link]

"Asymptotic expansion formulas for diffusion processes based on the perturbation method"
E. Guidotti, N. Yoshida
In preparation.

CONFERENCES AND SEMINARS

2026
Seminar at Stockholm Business School, Finance Section - *Presenter*
Swiss Finance Institute (SFI) PhD Workshop - *Session Chair*
Belgium–Switzerland AI Networking Event - *Presenter*
Finance Research Revolution (FRR) Conference - *Organizer*
Seminar at the University of Maryland, Finance Department - *Presenter*
Seminar at Linköping University, Institute of Statistics and Mathematical Statistics - *Presenter*
Volatility and Liquidity Workshop of the University of Pavia - *Presenter*
USI Event on SNSF Career Funding Opportunities - *Presenter*

2025
Conference on Neural Information Processing Systems (NeurIPS) - *Presentation by co-author*
Swiss Finance Institute (SFI) PhD Workshop - *Plenary Speaker*
Annual Meeting of the European Finance Association (EFA) - *Discussant*
Finance Research Revolution (FRR) Inaugural Conference - *Organizer*
Annual Meeting of the Swiss Society for Financial Market Research (SGF) - *Discussant*

Workshop on Numerical Methods For Finance and Insurance (NMFIW) - *Presenter*

2024 (*presenter*)
International Conference on Computational and Methodological Statistics (CMStatistics)
Institute for Mathematical Statistics - Asia-Pacific Rim Meeting (IMS-APRM)

2023 (*presenter*)
Seminar at EPFL, Swiss Finance Institute
Seminar at Linköping University, Machine Learning Series (IDA)
Seminar at OST, Interdisciplinary Center for Artificial Intelligence (ICAI)
Seminar at University of Milan, Information Systems and Knowledge Management (ISLab)
International Conference on Quantitative Finance and Financial Econometrics (QFFE)
Doctoral Workshop of the German Finance Association (DGF)

2022 (*presenter*)
Conference on Neural Information Processing Systems (NeurIPS)
Annual Meeting of the Financial Management Association (FMA)
Annual Meeting of the European Finance Association (EFA)
International Conference on Quantitative Finance and Financial Econometrics (QFFE)
Seminar at HEC Montréal, Department of Finance

2021 (*presenter*)
International Conference on Computational and Methodological Statistics (CMStatistics)
Annual Financial Market Liquidity Conference (AFML)
International Risk Management Conference (IRMC)
Annual Meeting of the German Finance Association (DGF)
IUPAP Conference on Computational Physics (CCP)
Quantact Seminar

2020 (*presenter*)
International Conference on Computational and Methodological Statistics (CMStatistics)
Next Einstein Forum Global Gathering
European R Users Meeting (eRum)

2019 (*presenter*)
International Conference on Computational and Methodological Statistics (CMStatistics)
European Meeting of Statisticians (EMS)

2018 (*presenter*)
Seminar at the University of Tokyo, Graduate School of Mathematical Sciences

2017 (*presenter*)
International Conference on Computational and Methodological Statistics (CMStatistics)
Annual Meeting of the Association for Mathematics Applied to Social and Economic Sciences
R/Finance Conference
Seminar at the University of Tokyo, Graduate School of Mathematical Sciences

RESEARCH SOFTWARE

"COVID-19 Data Hub"
Total downloads: 260,000+. Data downloads: 6,230,000+. Links: [GitHub] [CRAN] [PyPI]

"bidask: Efficient Estimation of Bid-Ask Spreads from Open, High, Low, and Close Prices"
Total downloads: 16,000+. Links: [GitHub] [CRAN] [PyPI]

"bornrule: Classification with Born's Rule"
Total downloads: 11,000+. Links [GitHub] [PyPI]

"torchabc: A simple abstract class for training and inference in PyTorch"
Total downloads: 42,000+. Links [GitHub] [PyPI]

"calculus: High-Dimensional Numerical and Symbolic Calculus in R"
Total downloads: 52,000+. Links: [GitHub] [CRAN]

"RblDataLicense: R Interface to Bloomberg Data License"
Total downloads: 52,000+. Links [GitHub] [CRAN]

"yuimaGUI: A Graphical User Interface for the yuima Package"
Total downloads: 29,000+. Links: [CRAN]

TEACHING

Fit for AI in Asset Management (Co-Teaching) — March 2027
— *Executive course*, University of St. Gallen

USI Workshop on NYSE Trade and Quote Data (Teaching) — September 2026
— *Workshop*, University of Lugano

Deploying Machine Learning Models on Google Cloud Platform (Teaching) — 2022 — 2027
— *Master* in Data Science for Economics, University of Milan

Corporate Finance (Teaching) — 2024 — 2025
— *Master* in Finance, University of Lugano

Stochastic Differential Equations with YUIMA (Co-Teaching) — March 2023
— *Ph.D.* course, Linköping University

Data Acquisition, Management, and Visualization (Teaching) — 2020 — 2022
— *Master* in Data Science for Economics, University of Milan

Coding for Data Science and Data Management (Co-Teaching) — 2019 — 2020
Text Mining and Sentiment Analysis (Co-Teaching)
— *Master* in Data Science for Economics, University of Milan

YUIMA Summer School on Computational and Statistical Methods for Stochastic Process (Co-Teaching) — June 2019
— *Ph.D.* course, Brixen-Bressanone, Italy

Asset Pricing (Teaching Assistant) — 2020 — 2023
Portfolio Management (Teaching Assistant)
Quantitative Methods for Finance (Teaching Assistant)
Research in Financial Analysis (Teaching Assistant)
— *Master* in Finance, University of Neuchâtel

Numerical and Statistical Methods for Finance (Teaching Assistant) 2018 — 2020
— *Master* in Data Science for Economics, University of Milan

Mathematics (Teaching Assistant) 2018 — 2020
— *Bachelor* in Political Science, University of Milan

GRANTS AND AWARDS

2025 — Swiss National Science Foundation — Ambizione (SNSF Grant #233173),
"*Financial Market Microstructure, Big Data, and AI",* 2025–2029 (CHF 607'360)
2025 — Open Data Quality Award by the Canadian Open Data Society
2024 — USI Research Grant in Open Research Data (CHF 50'000)
2024 — R Consortium ISC Grant Program
2022 — Neural Information Processing Systems (NeurIPS) Scholar Award
2022 — Mobility Scholarship of the University of Neuchâtel
2022 — Fonds Gold Open Access of the University of Neuchâtel
2021 — Google Cloud Research Credits
2020 — R Consortium ISC Grant Program
2020 — Institute for Data Valorization (IVADO) Research Grant

ACADEMIC SERVICE

R*eviewer of peer-reviewed journals and conferences*:

Advances in Neural Information Processing Systems (2026)
Journal of Business & Economic Statistics (2026)
Journal of Statistical Software (2025)
Journal of Behavioral and Experimental Finance (2025)
Mathematical Finance (2025)
Journal of Computational and Applied Mathematics (2025)
Advances in Neural Information Processing Systems (2025)
Journal of Financial Econometrics (2024)
Financial Markets and Portfolio Management (2024)
The R Journal (2023)
Scientific Data, Nature (2022)

Jury member of PhD theses:

"Advancing Portfolio Decision Making with Machine Learning: Attention Mechanism, Convolutional Neural Networks, and Neural Tangent Kernels"
Swiss Finance Institute PhD in Finance - École Polytechnique Fédérale de Lausanne (2025)
Candidate: Boris Kuznetsov
Supervisor: Semyon Malamud
Jury members: Daniel Kuhn, Damir Filipovic, Zhang Yuan, Emanuele Guidotti

"Empirical Asset Pricing: the Role of Big Data, Machine Learning, and Deep Learning"
Swiss Finance Institute PhD in Finance - École Polytechnique Fédérale de Lausanne (2024)
Candidate: Teng Andrea Xu
Supervisor: Semyon Malamud
Jury members: Daniel Kuhn, Damir Filipovic, Zhang Yuan, Emanuele Guidotti

Supervision of Master theses:

"Multi-Agent Reinforcement Learning for a Model-based Limit Order Book Simulation"
Master's Degree in Data Science and Economics, University of Milan (2024)
Candidate: Shihab Hamati
Supervisor: Nicolò Cesa-Bianchi
Co-supervisors: Alfio Ferrara, Emanuele Guidotti

“Image Classification with Born’s Rule: Assessing Adaptability, Compatibility, and Performance of the Text Classification Algorithm in the Image Domain”
Master's Degree in Data Science and Economics, University of Milan (2024)
Candidate: Guido Giacomo Mussini
Supervisor: Alfio Ferrara
Co-supervisor: Emanuele Guidotti

Research Assistants:

Viktoriia Tantsiura

OTHER EMPLOYMENT AND ADDITIONAL AFFILIATIONS

Japan Science and Technology Agency CREST 2017 — present

Partner at Algo Finance Sagl 2017 — 2025

PROGRAMMING

Proficient in Python, R, SQL, C++ / Shell scripting / Cloud computing / WRDS HPC

LANGUAGES

Italian (native), English (C1).